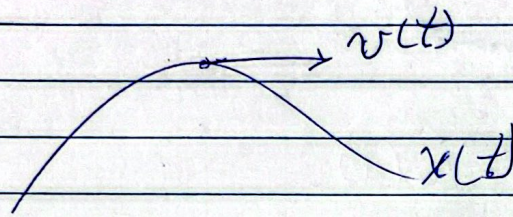


Problema #2

$$\vec{v}(t) = \hat{i} + 4t\hat{j} - 3t^2\hat{k}$$

$$t = 1.$$

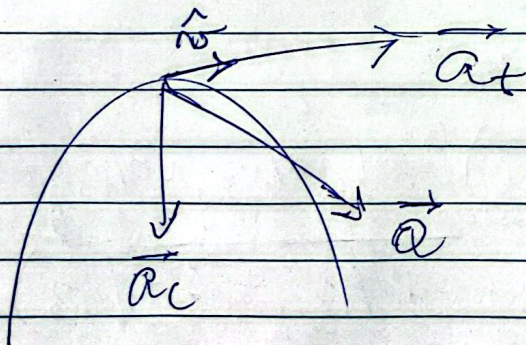
- ① El vector unitario en la dirección de la tangente de la trayectoria



$$\hat{v}(1) = \frac{\vec{v}(1)}{|\vec{v}(1)|} = \frac{(1, 4, -3)}{(1^2 + 4^2 + 3^2)^{1/2}}$$

$$\hat{v}(1) = \frac{1}{\sqrt{26}} (1, 4, -3)$$

- ② El vector aceleración tangencial



$$\vec{a} = \dot{\vec{v}} = (1, 4t, -3t^2)'$$

$$= (0, 4, -6t)$$

$$\vec{a}(1) = (0, 4, -6)$$

$$\vec{a}(1) \cdot \hat{v}(1) = a_t$$

$$a_t = (0, 4, -6) \cdot (1, 4, -3) \frac{1}{\sqrt{26}}$$

$$= (0 + 16 + 18) \frac{1}{\sqrt{26}}$$

$$a_t = \frac{17\sqrt{26}}{13}$$

$$\vec{a}_t = a_t \cdot \hat{v}(1)$$

$$= \frac{17\sqrt{26}}{13} (1, 4, -3) \cdot \frac{1}{\sqrt{26}}$$

$$\vec{a}_t = \left(\frac{17}{13}, \frac{68}{13}, \frac{-51}{13} \right)$$

③ El vector aceleración normal

$$\vec{a} = \vec{a}_t + \vec{a}_n$$

$$\vec{a}_n = \vec{a} - \vec{a}_t$$

$$= (0, 4, -6) - \left(\frac{17}{13}, \frac{68}{13}, \frac{-51}{13} \right)$$

$$\vec{a}_n = \left(-\frac{17}{13}, -\frac{16}{13}, -\frac{27}{13} \right)$$

- ④ El vector unitario en la dirección de la normal de la trayectoria

$$\hat{a}_c = \frac{\vec{a}_c}{|\vec{a}_c|} = \frac{(-17/13, -16/13, -27/13)}{\left[\left(\frac{17}{13}\right)^2 + \left(\frac{16}{13}\right)^2 + \left(\frac{27}{13}\right)^2 \right]^{1/2}}$$

$$\boxed{\hat{a}_c = \frac{1}{7\sqrt{26}} (-17, -16, -27)}$$

- ⑤ El radio de curvatura ρ .

$$a_c = \frac{v^2}{\rho}$$

$$\rho = \frac{v^2}{a_c} = \frac{(\vec{v}(1) \cdot \vec{v}(1))}{[\vec{a}_c(1) \cdot \vec{a}_c(1)]^{1/2}}$$

$$= \frac{1^2 + 4^2 + 3^2}{\left[\left(\frac{17}{13}\right)^2 + \left(\frac{16}{13}\right)^2 + \left(\frac{27}{13}\right)^2 \right]^{1/2}}$$

$$\left[\left(\frac{17}{13}\right)^2 + \left(\frac{16}{13}\right)^2 + \left(\frac{27}{13}\right)^2 \right]^{1/2}$$

$$\boxed{\rho = \frac{13\sqrt{26}}{7}}$$